Certified Reference Material (CRM) Certificate

1. CRM Identification

CRM Name: Certified Reference Material AS007

CRM Code: CRM AS 007

Batch Number: 1

Date of Certification: 21-10-2025

Expiry Date: One year after opening

Produced and certified by: Azuresky Pty Ltd, South Africa

2. Intended Use

This CRM is intended for calibration, validation, control sample and training of staff for the analytical methods listed in section 6 of this document.

3. Sample Description

Coal sample, ±150g per unit, air-dried, pulverised to <212 µm.

4. Instructions for Use

- Equilibrate to room temperature before use (conditioning), for at least 3 hours.

- Determine inherent moisture and adjust dry-basis values to air-dry if required.

- Mix the material thoroughly before weighing.

- Use within 12 months of opening.

- Store in a room not exceeding 23°C.

5. Homogeneity Assessment

Ten units were randomly selected and analyzed for quick ash content. Homogeneity was evaluated using ANOVA. No significant between-bottle variation was observed ($F_{calc} < F_{crit}$, $p = 0.05$). This confirms sufficient homogeneity for intended use.

6. Certified Values – Dry Basis

Measurand	Assigned Value	Uncertainty (±)	Unit	Method
ISO Ash	19.0	±0.46	%	ISO 1171
ISO Volatile Matter	26.8	±0.35	%	ISO 562
Calorific Value	25.72	± 0.16	MJ/kg	ISO 1928
Total Sulphur	0.97	±0.03	%	ASTM D4239
Quick Ash	19.2	±0.47	%	Internal Method

7. Indicative Values – Air-Dry Basis (to be used as a guideline)

The average moisture obtained from 22 laboratories is 3.70 %

Measurand	Value	Tolerance	Unit	Minimum	Maximum	Method
ISO Ash	18.3	0.55	%	17.8	18.9	ISO 1171
ISO Volatile Matter	25.8	1.03	%	24.8	26.8	ISO 562
Calorific Value	24.77	0.30	MJ/kg	24.47	25.07	ISO 1928
Total Sulphur	0.93	0.10	%	0.83	1.03	ASTM D4239
Quick Ash	18.5	0.56	%	17.9	19.1	Internal Method

Tolerance values are based on the following:

- ISO 1171 reproducibility (ISO Ash): 3% of the mean for ash values > 10%

- ISO 1928 reproducibility (Calorific value): 0.30 MJ/kg

- ISO 562 reproducibility (Volatile matter): 0.5% absolute or 4% of the mean, whichever is greater

- ASTM D4239 reproducibility (Total Sulphur): $(0.02+0.09\bar{x})$

8. Stability Statement

At the time of certification, full long-term stability validation of this CRM is ongoing.
Preliminary evaluations and material characteristics suggest stability under recommended storage conditions. As a precautionary guideline, users are advised to store the CRM at 18- 23 °C in a sealed container and use the material within one year from the date of opening. This provisional guidance will be updated upon completion of the full stability study.

9. Analysis and Statistical Validity

All certified values were derived from replicate measurements conducted by 22 laboratories operating under ISO/IEC 17025-compliant quality systems. Assigned values are calculated as the consensus means. Uncertainty values represent expanded uncertainty (k=2) at 95% CL.

10. Authorization

Authorized by: Nonhlanhla Msibi
Managing Director, Azuresky Pty Ltd Date: 22-10-2025
Signature: _____
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